FINANCIAL SALES PERFORMANCE ANALYSIS



BY

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INTRODUCTION

This report analyzes a financial sales dataset using Python and Power BI to understand sales and business performance. The main objectives are to analyze sales and profit, compare products and states, study customer segments, and identify monthly sales trends using data analysis and visualizations.

The analysis helps identify top-performing products and customer segments and understand sales and profit patterns. The report presents insights based on sales, profit, units sold, discounts, products, states, and customer segments. This project shows how data analytics can turn raw data into useful insights and support better business decisions.

AIM

The main aim of this project is to analyze the financial sales dataset using Python, Exploratory Data Analysis (EDA), and Power BI to understand sales and business performance. The analysis focuses on sales, profit, products, states, customer segments, and monthly sales trends. Specifically, the project aims to:

1. **Analyze Sales and Profit:** Examine sales and profit to understand overall business performance.
2. **Compare Products and States:** Identify top-performing products and states based on sales and profit.
3. **Study Customer Segments and Trends:** Analyze customer segments, monthly sales patterns, units sold, and discounts.
4. **Present Business Insights:** Develop an interactive Power BI dashboard to display key findings and support better business decisions.

This project demonstrates how Python, EDA, and Power BI can be used to analyze sales data, identify important patterns, and generate useful business insights.

OBJECTIVES

1. Data Collection:

The financial sales dataset contains information about sales, profit, products, states, customer segments, units sold, and discounts. This data provides a foundation for analyzing business performance and identifying sales trends.

2. Data Cleaning and Preprocessing:

The dataset was cleaned and prepared using Python to improve data quality and consistency. The process included checking missing values, duplicate records, data types, and incorrect values to make the data suitable for analysis.

3. Exploratory Data Analysis (EDA):

EDA was performed using Python to identify patterns and trends in sales and profit. The analysis examined product performance, state-wise sales, customer segments, discounts, units sold, and monthly sales patterns.

4. Sales and Profitability Analysis:

The analysis compares sales and profit across different products, states, and customer segments. It also examines units sold and discounts to identify high-performing products, profitable segments, and areas for improvement.

5. Data Visualization and Dashboard Development:

The prepared dataset was used in Power BI to develop an interactive dashboard. Charts, key performance indicators (KPIs), and filters were used to present sales, profit, product performance, and customer segment insights.

6. Business Insights and Decision-Making:

The project aims to identify useful insights from sales data by analyzing business performance, product trends, and customer segments.

DATA OVERVIEW

Column name	Data Type
Segment	Object
State	Object
Product	Object
Units Sold	int64
Manufacturing Price	float64
Sale Price	float64
Discounts	float64
Sales	float64
COGS	float64
Profit	float64
Date	datetime64[ns]
Month Number	float64
Month Name	Object
Year	int64
Profit Margin (%)	float64
Discount Rate (%)	float64

DATA ANALYSIS

1. Descriptive Analysis

This section provides an overview of the financial sales dataset by examining total sales, profit, units sold, and discounts. It also analyzes products, states, and customer segments to understand overall business performance. Statistical summaries are used to understand the distribution of sales and profit values. This analysis provides a clear picture of the dataset and helps identify important business information.

2. Sales Relationship and Trend Analysis

This section examines sales and profit patterns across different products, states, and customer segments. It also analyzes monthly sales trends to identify changes in business performance over time. The analysis helps identify periods of high and low sales and understand variations in revenue. These comparisons provide insights into sales performance and business trends.

3. Product and Customer Segment Analysis

This section compares sales and profit across different products and customer segments. It helps identify popular products, high-performing segments, and differences in purchasing patterns. The analysis also examines the contribution of different products and customer groups to overall sales. These findings help understand customer demand and identify areas with better sales opportunities.

4. Profitability and Performance Analysis

This section focuses on evaluating sales and profit across products, states, and customer segments. It also examines units sold and discounts to identify profitable areas and opportunities for business improvement. Comparing sales and profit helps identify products and regions that perform well and those that may require attention. The results provide a better understanding of business profitability and the factors associated with profit variations.

5. Data Visualization and Dashboard Analysis

This section presents key findings through an interactive Power BI dashboard using charts, key performance indicators (KPIs), and filters. The dashboard helps users explore sales performance, profit distribution, product trends, and regional results to support better business decisions. Interactive features allow users to compare different categories and view specific information easily. The visualizations present complex sales data in a simple and understandable format.

QUESTIONS

1. **Which product generates the highest total sales?**

   ```
   df.groupby('Product')['Sales'].sum().sort_values(ascending=False)
   ```

2. **Which country generates the highest total sales?**

   ```
   df.groupby('State')['Sales'].sum().sort_values(ascending=False)
   ```

3. **Which product has the highest average profit margin?**

   ```
   df['Profit Margin (%)'] = (df['Profit'] / df['Sales']) * 100


   df.groupby('Product')['Profit Margin (%)'].sum().sort_values(ascending=False)
   ```

4. **Does the discount rate affect profit margin?**

   ```
   df['Discount Rate (%)'] = (df['Discounts'] / df['Sales']) * 100
   ```

```
print(df[['Discount Rate (%)', 'Profit Margin (%)']].corr())
sns.scatterplot(
    data=df,
    x='Discount Rate (%)',
    y='Profit Margin (%)'
)
plt.title('Discount Rate vs Profit Margin')
plt.show()
```

5. **Is there a relationship between discounts and profit?**

```
print(df[['Discounts', 'Profit']].corr())

sns.scatterplot(data=df, x='Discounts', y='Profit')
plt.title('Discounts vs Profit')
plt.show()
```

6. **Which products receive the highest average discount?**

```
df.groupby('Product')["Discounts"].mean().sort_values(ascending=False)
```

7. **Which products experienced the largest change in sales between 2013 and 2014?**

```
yearly_sales = df.pivot_table(
    index='Product',
    columns='Year',
    values='Sales',
    aggfunc='sum'
)
yearly_sales['Sales_Change'] = (
yearly_sales[2014] - yearly_sales[2013]
)


print(yearly_sales.sort_values('Sales_Change', ascending=False))
```

8. **Which segment contributes the largest share of total revenue?**

```
segment_sales = df.groupby('Segment')['Sales'].sum()


segment_contribution = (segment_sales / df['Sales'].sum()) * 100


print(segment_contribution.sort_values(ascending=False))
```

9. **Which products have manufacturing costs higher than the overall average?**

```
average_manufacturing_price = df['Manufacturing Price'].mean()

product_manufacturing = df.groupby('Product')['Manufacturing Price'].mean()

print(product_manufacturing[
   product_manufacturing > average_manufacturing_price])
```

10. **Which segment sells the highest number of units?**

```
segment_units = df.groupby('Segment')['Units Sold'].sum().sort_values(ascending=False)


print(segment_units)
print("Highest-selling segment:", segment_units.idxmax())
print("Total units sold:", segment_units.max())
```

11. **How do sales and profit differ between customer segments?**

```
segment_comparison = df.groupby('Segment').agg(
    Total_Sales=('Sales', 'sum'),
    Total_Profit=('Profit', 'sum'),
    Average_Sales=('Sales', 'mean'),
    Average_Profit=('Profit', 'mean')
)
```

```
print(segment_comparison.sort_values('Total
_Sales', ascending=False))
```

12. Which country has the highest ratio of profit to COGS?

```
country_efficiency = df.groupby('State').agg(
    Total_Profit=('Profit', 'sum'),
    Total_COGS=('COGS', 'sum')
)

country_efficiency['Profit to COGS (%)'] = (
    country_efficiency['Total_Profit'] /
    country_efficiency['Total_COGS'].replace(0,
float('nan'))
) * 100

print(country_efficiency.sort_values('Profit to
COGS (%)', ascending=False))
```

13. **What is the relationship between units sold and total sales?**

```
print(df[['Units Sold', 'Sales']].corr())
```

14. **How do sales change over time?**

```
df['Date'] = pd.to_datetime(df['Date'], errors='coerce', dayfirst=True)

monthly_sales = (
    df.dropna(subset=['Date'])
    .set_index('Date')['Sales']
    .resample('ME')
    .sum()
)

monthly_sales.plot(kind='line', marker='o', figsize=(12, 5))

plt.title('Monthly Sales Trend')

plt.xlabel('Month')

plt.ylabel('Total Sales')

plt.tight_layout()
```

```
plt.show()
```

15. Which customer segment contributes the highest percentage of sales?

```
segment_sales =
df.groupby('Segment')['Sales'].sum()
plt.figure(figsize=(7, 7))
plt.pie(
    segment_sales,
    labels=segment_sales.index,
    autopct='%1.1f%%',
    startangle=90
)
plt.title('Sales Contribution by Segment')
plt.show()
```

16. How does the average selling price compare with the manufacturing price for each product?

```
price_comparison = df.groupby('Product')[
    ['Sale Price', 'Manufacturing Price']
```

```
].mean()

price_comparison.plot(kind='bar', figsize=(12, 6))

plt.title('Average Selling Price vs Manufacturing Price')

plt.xlabel('Product')

plt.ylabel('Average Price')

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()
```

CONCLUSION

The analysis of the financial sales dataset provides useful insights into sales performance, profitability, and customer segments. The project examines important factors such as sales, profit, units sold, discounts, products, states, and monthly sales patterns. Using Python, the data was analyzed through statistical summaries, comparisons, filtering, and visualizations. These analyses help identify top-performing products, compare sales across states and customer segments, and understand variations in sales and profit.

Overall, this project demonstrates how data analytics can transform raw sales data into meaningful business insights. The analysis helps identify profitable products, understand customer segments, and explore sales trends across different regions and time periods. The interactive Power BI dashboard presents these findings through charts, KPIs, and filters, making

the results easier to understand. These insights can support better business decisions, improve sales strategies, and help businesses identify opportunities for growth and profitability.

1. **Which product generates the highest total sales?**

   The groupby() and sum() functions were used to calculate the total sales generated by each product. The results were sorted in descending order to identify the products with the highest sales. This analysis helps compare product performance and identify which products contribute the most to overall sales.

Product	
Smart Tv	329445.13
Tablet	291401.28
Laptop	240240.06
Mobile	230912.32
Refrigerator	228690.84
Washing Machine	196095.90

2. Which country generates the highest total sales?

The groupby() and sum() functions were used to calculate the total sales generated by each state. The results were sorted in descending order to identify the state with the highest sales. This analysis helps compare sales performance across states and identify the strongest-performing region. Goa generates the highest total sales of 324,507.32.

State	
Goa	324507.32
Kerala	318000.57
Uttar Pradesh	310246.26
Mizoram	303224.82
Tamil Nadu	260806.56

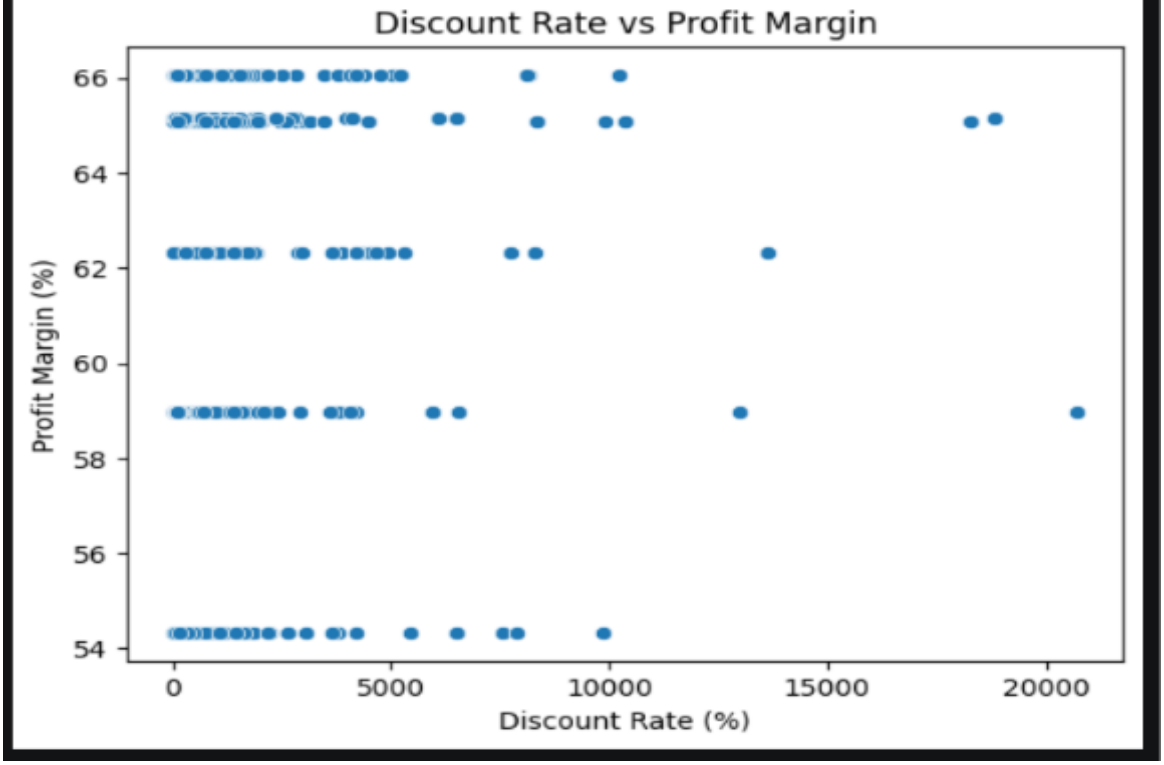
3.Which product has the highest average profit margin?

The groupby() and sum() functions were used to calculate the total profit margin percentage for each product. The results were sorted in descending order to identify the product with the highest total profit margin percentage. This analysis helps compare profitability across different products and identify which products contribute the most to the overall profit margin. Smart TV has the highest total profit margin percentage, with approximately 8,730.23%.

Product	
Smart Tv	8730.234136
Tablet	7914.691092
Refrigerator	7332.265446
Laptop	7224.019025
Washing Machine	6310.256410
Mobile	5976.610429

4.Does the discount rate affect profit margin?

	Discount Rate (%)	Profit Margin (%)
Discount Rate (%)	1.000000	-0.002798
Profit Margin (%)	-0.002798	1.000000

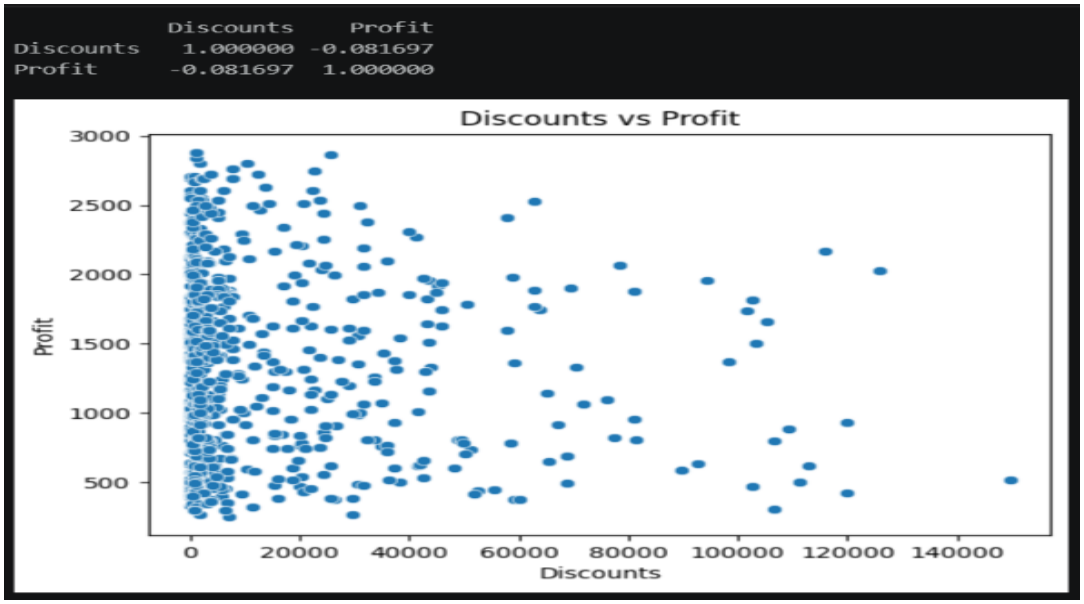


The corr() function was used to identify the relationship between the Discount Rate (%) and Profit Margin (%). The correlation value is approximately -0.0028, indicating an almost zero negative correlation between the two variables. The scatter plot also shows no clear relationship between discount rate and profit margin. Therefore, based on this analysis, the discount rate does not appear to have a

meaningful linear relationship with profit margin in this dataset.

5. **Is there a relationship between discounts and profit?**

The corr() function was used to identify the relationship between Discounts and Profit. The correlation value is approximately -0.0817, indicating a very weak negative correlation between the two variables. The scatter plot shows that profit varies considerably at different discount amounts, with no strong relationship between them. Therefore, discounts have a very weak negative linear relationship with profit in this dataset.



6.Which products receive the highest average discount?

The groupby() and mean() functions were used to calculate the average discounts for each product, and sort_values(ascending=False) arranged the results in descending order. The analysis shows that Refrigerators receive the highest average discount of 14,214.63, followed by Washing Machines at 13,659.98 and Laptops at 13,617.33. Smart TVs have the lowest average discount at 11,867.20. This analysis helps identify which products receive higher discounts.

Product	
Refrigerator	14214.626126
Washing Machine	13659.984953
Laptop	13617.327838
Tablet	13531.707953
Mobile	12232.286909
Smart Tv	11867.197910

7. **Which products experienced the largest change in sales between 2013 and 2014?**

   The pivot_table() function was used to calculate total sales for each product in 2013 and 2014. The difference between the two years was calculated using Sales_Change and sorted in descending order. The analysis shows that Tablets experienced the largest increase in sales, rising by 174,528.96, followed by Smart TVs with an increase of 164,413.13 and Laptops with 146,081.70. This analysis helps identify products with the strongest sales growth between 2013 and 2014.

8. **Which segment contributes the largest share of total revenue?**

   The groupby() and sum() functions were used to calculate total sales for each customer segment. The percentage contribution of each segment to total revenue was then calculated and sorted in descending order. The analysis shows that Wholesale contributes the largest

share of total revenue at 35.25%, followed by Online at 33.24% and Retail at 31.51%. This analysis helps identify the customer segment that contributes the most to overall sales.

Segment	
Wholesale	35.245428
Online	33.239956
Retail	31.514616

9. **Which products have manufacturing costs higher than the overall average?**

The mean() function was used to calculate the overall average Manufacturing Price, while groupby() and mean() calculated the average manufacturing price for each product. The results were then filtered to display only products with values higher than the overall average. The analysis shows that Mobile (11.91), Tablet (10.49), and Smart TV (10.27) have manufacturing costs higher than the overall average. This analysis helps identify

products with relatively higher manufacturing costs.

Product	
Mobile	11.91
Smart Tv	10.27
Tablet	10.49

10. Which segment sells the highest number of units?

The groupby() and sum() functions were used to calculate the total units sold by each segment, and sort_values(ascending=False) arranged the results in descending order. The analysis shows that Wholesale is the highest-selling segment, with 20,269 units sold, followed by Online with 19,277 units and Retail with 18,340 units. The idxmax() function identifies Wholesale as the segment with the highest total units sold. This analysis helps identify the segment that generates the highest sales volume.

Segment	
Wholesale	20269

Online	19277
Retail	18340

Highest-selling segment: Wholesale

Total units sold: 20269

11. How do sales and profit differ between customer segments?

The groupby() and agg() functions were used to compare total sales, total profit, average sales, and average profit across the three segments. The results were sorted by total sales in descending order. The analysis shows that Wholesale has the highest total sales (534,597.55) and total profit (334,833.97). Online has the highest average sales (2,271.08) and average profit (1,414.38) per record, while Retail has the lowest total sales and profit. This analysis helps compare segment performance and identify differences in overall and average profitability.

12. Which country has the highest ratio of profit to COGS?

The groupby() and agg() functions were used to calculate the total profit and total Cost of Goods Sold (COGS) for each state. The profit-to-COGS percentage was then calculated and sorted in descending order. The analysis shows that Kerala has the highest profit-to-COGS ratio at 168.01%, followed by Tamil Nadu at 167.59% and Goa at 164.56%. Mizoram has the lowest ratio among the listed states at 162.73%. This analysis helps compare profitability relative to the cost of goods sold across different states.

13. What is the relationship between units sold and total sales?

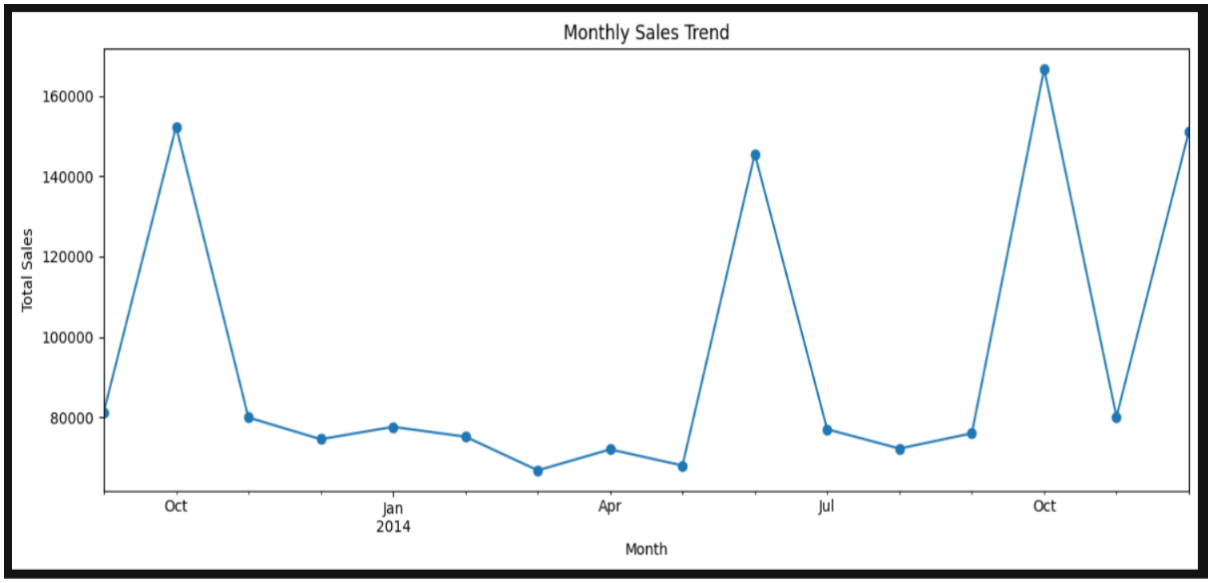
The corr() function was used to identify the relationship between Units Sold and Sales. The correlation coefficient is 0.9747, indicating a very strong positive correlation between the two variables. This means that

higher numbers of units sold are generally associated with higher total sales. This analysis helps understand how sales volume is related to revenue generation.

Units Sold	Sales	
Units Sold	1.000000	0.974743
Sales	0.974743	1.000000

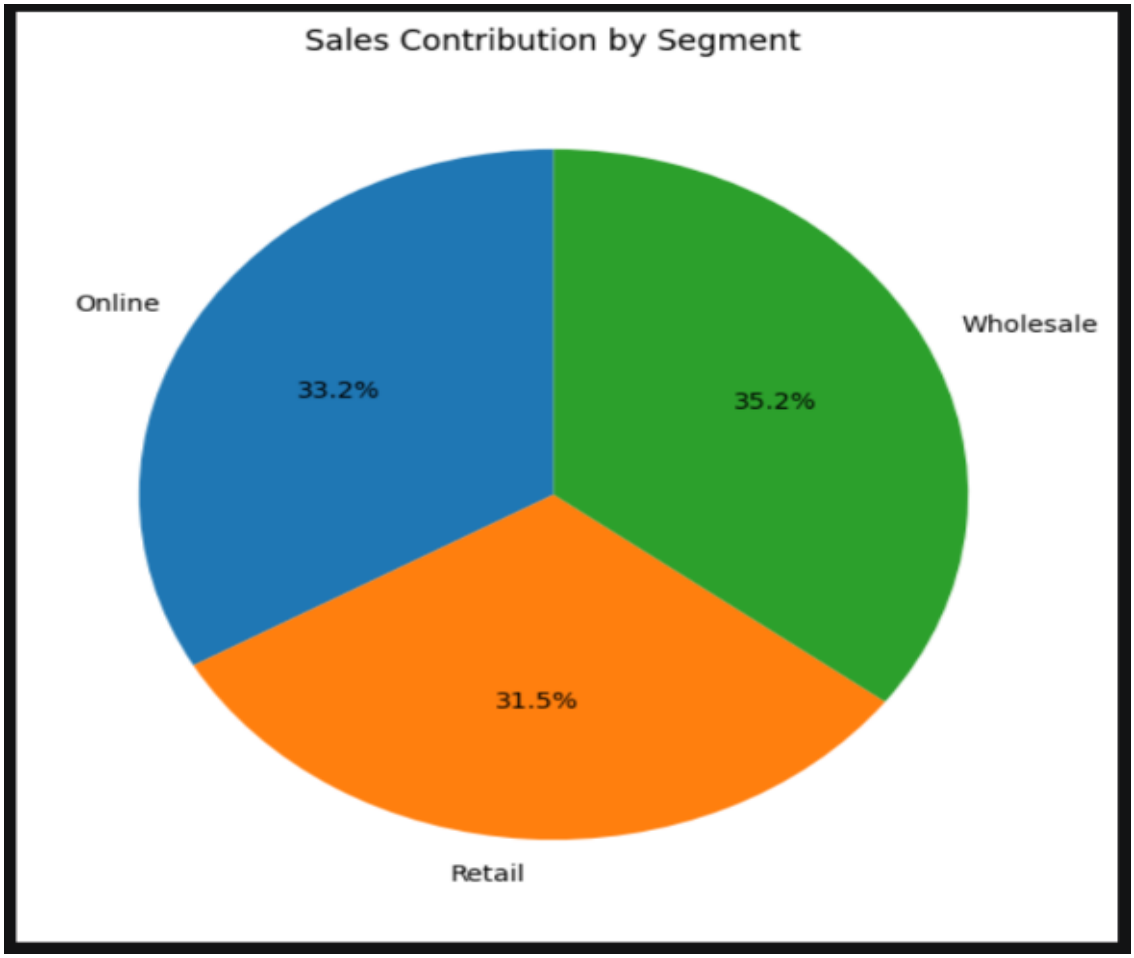
14. How do sales change over time?

The resample('ME').sum() method was used to calculate total sales for each month, and a line chart was created to visualize sales trends over time. The analysis shows that monthly sales fluctuate significantly, with major peaks around October 2013, June 2014, October 2014, and December 2014. Sales are relatively lower during several months between these peaks. October 2014 records the highest sales in the chart, at approximately 167,000. This analysis helps identify high-performing months and understand changes in sales over time.



15. Which customer segment contributes the highest percentage of sales?

The groupby() and sum() functions were used to calculate total sales for each customer segment, and a pie chart was created to display their percentage contributions. The analysis shows that Wholesale contributes the highest percentage of total sales at 35.2%, followed by Online at 33.2% and Retail at 31.5%. This analysis helps identify the customer segment that contributes the largest share of overall sales.



16. How does the average selling price compare with the manufacturing price for each product?

The groupby() and mean() functions were used to calculate the average selling price and manufacturing price for each product. A grouped bar chart was created to compare the two prices. The analysis shows that the average selling price is higher than the manufacturing price for all products. Smart TVs have the highest average

selling price at approximately 29.5, while Washing Machines have the lowest at around 21.5. Mobile phones have the highest average manufacturing price at approximately 11.9. This analysis helps compare product pricing and manufacturing costs.

